STAT GU4204 Midterm Cheat Sheet

Key Definitions

Statistical inference: Procedure producing a probabilistic statement about parts of a statistical model.

Parameter: Characteristic(s) determining the joint distribution of the r.v.'s of interest.

Parameter space Ω: Set of all possible values of θ.

Statistic: $T = \varphi(X_1, \ldots, X_n)$ — function of data ONLY, no unknown parameters.

Estimator: $\hat{\theta}_n = \varphi(X_1, \ldots, X_n)$ — a statistic used to estimate a parameter; it is a *random variable*.

Estimate: $\varphi(x_1, \ldots, x_n)$ — the realized *number* after plugging in observed data.

Bias: $b(T_n) = E(T_n) - g(\theta)$.

Unbiased: $E(T_n) = g(\theta)$ for ALL $\theta \in \Omega$ (bias $= 0$ everywhere).

Consistent: $\hat{\theta}_n \xrightarrow{P} \theta$ as $n \to \infty$. Large-sample property; does NOT tell us about finite-n performance.

MSE: $E[(T_n - g(\theta))^2] = \mathrm{Var}(T_n) + [\mathrm{bias}(T_n)]^2$.

Dominates: T_n dominates S_n if $\mathrm{MSE}(T_n) \le \mathrm{MSE}(S_n)$ for ALL θ.

Inadmissible: S_n is inadmissible if some T_n dominates it.

MVUE: Minimum Variance Unbiased Estimator — unbiased AND has smallest variance among all unbiased estimators.

Efficient estimator: Unbiased estimator that achieves the Cramér-Rao lower bound.

Sufficient statistic: T is sufficient for θ if the conditional distribution of $X|T{=}t$ does NOT depend on θ.

Likelihood: $L(\theta) = \prod f(X_i, \theta)$.

Log-likelihood: $\ell(\theta) = \sum \log f(X_i, \theta)$.

Score function: $\dot{\ell}(x, \theta) = \frac{\partial}{\partial\theta} \log f(x, \theta)$.

Score equation: $\sum \dot{\ell}(X_i, \theta) = 0$ (solve for MLE).

Fisher information: $I(\theta) = E[\dot{\ell}^2(X, \theta)] = \mathrm{Var}[\dot{\ell}(X, \theta)]$. Score has mean 0.

Sampling distribution: Distribution of a statistic $T = \varphi(X_1, \ldots, X_n)$.

Pivot: $\Psi(\mathbf{X}_n, g(\theta))$ — function of data AND parameter whose distribution does NOT depend on θ.

Confidence interval: Random interval (A, B) such that $P_\theta(A \le g(\theta) \le B) \ge 1 - \alpha$ for all θ. **Exact** if equality holds.

Prior $\xi(\theta)$: Belief about θ BEFORE observing data.

Posterior $\xi(\theta|x)$: Updated belief about θ AFTER observing data.

Conjugate prior: Prior such that posterior belongs to the SAME family as the prior.

Bayes estimator: Minimizes expected posterior loss.

- Squared error loss $\Rightarrow$ posterior mean
- Absolute error loss $\Rightarrow$ posterior median

Distributions Reference

Dist	PDF/PMF	$E(X)$	$\mathrm{Var}(X)$	MGF
$\mathrm{Ber}(p)$	$p^x(1-p)^{1-x}$	p	$p(1-p)$	$1-p+pe^t$
$\mathrm{Bin}(n,p)$	$\binom{n}{k}p^k(1-p)^{n-k}$	np	$np(1-p)$	$(1-p+pe^t)^n$
$\mathrm{Poi}(\lambda)$	$e^{-\lambda}\lambda^k/k!$	λ	λ	$e^{\lambda(e^t-1)}$
$\mathrm{Exp}(\theta)$	$\theta e^{-\theta x}$	$1/\theta$	$1/\theta^2$	$\theta/(\theta-t)$
$N(\mu,\sigma^2)$	$\frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2	$e^{\mu t+\sigma^2t^2/2}$
$\mathrm{Unif}[a,b]$	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{tb}-e^{ta}}{t(b-a)}$
$\mathrm{Gam}(\alpha,\beta)$	$\frac{\beta^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-\beta x}$	α/β	α/β^2	$\left(\frac{\beta}{\beta-t}\right)^\alpha$
χ^2_d	$= \mathrm{Gam}(d/2, 1/2)$	d	$2d$	$(1-2t)^{-d/2}$
t_n	symm, heavy tails	0	$\frac{n}{n-2}$	—
$\mathrm{Beta}(\alpha,\beta)$	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	—

Gamma Function

$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1}e^{-x}\,dx$

$\Gamma(\alpha{+}1) = \alpha\,\Gamma(\alpha) \qquad \Gamma(n) = (n{-}1)! \qquad \Gamma(1/2) = \sqrt{\pi}$

Reproductive Properties (Indep. Sums)

X_i iid from	$\sum X_i \sim$
$\mathrm{Ber}(p)$	$\mathrm{Bin}(n,p)$
$\mathrm{Bin}(n_i,p)$	$\mathrm{Bin}(\sum n_i, p)$
$\mathrm{Gam}(\alpha_i,\beta)$ same β	$\mathrm{Gam}(\sum\alpha_i,\beta)$
$\chi^2_{m_i}$	$\chi^2_{\sum m_i}$
$\mathrm{Poi}(\lambda_i)$	$\mathrm{Poi}(\sum\lambda_i)$
$N(\mu_i,\sigma_i^2)$	$N(\sum\mu_i, \sum\sigma_i^2)$

Normal Properties

$X \sim N(\mu,\sigma^2) \Rightarrow aX + b \sim N(a\mu + b,\ a^2\sigma^2)$

$\bar{X}_n \sim N(\mu, \sigma^2/n)$ for iid normal sample.

CDF: $\Phi(z)$, symmetry: $1 - \Phi(x) = \Phi(-x)$

Key values: $z_{0.025} = 1.96$, $z_{0.05} = 1.645$, $z_{0.005} = 2.576$

Probability Inequalities

Markov: $W \ge 0 \ \Rightarrow\ P(W \ge t) \le E(W)/t$

Chebyshev: $P(|Z - EZ| \ge \tau) \le \mathrm{Var}(Z)/\tau^2$

On $\bar{X}$: $P(|\bar{X} - \mu| \ge \tau) \le \sigma^2/(n\tau^2)$

Sample size: want $P(|\bar{X} - \mu| \ge \varepsilon) \le \delta$, need $n \ge \sigma^2/(\varepsilon^2\delta)$.

Convergence

In probability ($Z_n \xrightarrow{P} Z$): $P(|Z_n - Z| > \varepsilon) \to 0$ for all $\varepsilon > 0$.

In distribution ($Z_n \xrightarrow{d} Z$): $F_n(u) \to F(u)$ at all continuity points of F.

Conv in prob $\Rightarrow$ conv in dist. Converse is FALSE in general (TRUE if limit is a constant).

WLLN: X_i iid, $E|X_1| < \infty \ \Rightarrow\ \bar{X}_n \xrightarrow{P} \mu$.

Conditions: iid, finite mean. Proof uses Chebyshev (if variance exists).

SLLN: $P(\lim \bar{X}_n = \mu) = 1$ (almost sure convergence, STRONGER).

CLT: X_i iid, mean μ, $\sigma^2 < \infty$:

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0,1)$$

Conditions: iid, finite variance. Gives APPROXIMATE dist for large n. Does NOT require normality of X_i.

Continuous Mapping Theorem:

- $Z_n \xrightarrow{P} b$, g cts at $b \Rightarrow g(Z_n) \xrightarrow{P} g(b)$
- $Z_n \xrightarrow{d} Z$, g continuous $\Rightarrow g(Z_n) \xrightarrow{d} g(Z)$

Slutsky: $X_n \xrightarrow{d} X$, $Y_n \xrightarrow{P} c$ (constant) $\Rightarrow X_nY_n \xrightarrow{d} cX$ and $X_n + Y_n \xrightarrow{d} X + c$.

Delta Method

If $\sqrt{n}(Z_n - \theta)/\sigma \xrightarrow{d} N(0,1)$ and $g'(\theta) \neq 0$:

$$\sqrt{n}\big(g(Z_n) - g(\theta)\big) \xrightarrow{d} N\big(0,\ \sigma^2[g'(\theta)]^2\big)$$

Requires $g'(\theta) \neq 0$. If $g'(\theta) = 0$, delta method breaks (need second-order).

Recipe: derivative of g, evaluate at θ, square it, multiply by original variance.

Method of Moments (MOM)

Set population moments = sample moments, solve:

1. $\mu_j(\theta) = E(X^j)$ for $j = 1, \ldots, k$ parameters
2. $\hat{\mu}_j = \frac{1}{n}\sum X_i^j$
3. Solve $\mu_j(\theta) = \hat{\mu}_j$

Dist	MOM Estimator
Ber(θ), Poi(λ)	$\hat{\theta} = \bar{X}$
$N(\mu, \sigma^2)$	$\hat{\mu} = \bar{X}$, $\hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$
Exp(λ)	$\hat{\lambda} = 1/\bar{X}$
Gam(α, β)	use $\bar{X} = \alpha/\beta$, $\hat{\mu}_2 - \bar{X}^2 = \alpha/\beta^2$

MOM is consistent (LLN + CMT). Not unique. Not always optimal.

Maximum Likelihood Estimation (MLE)

$L(\theta) = \prod f(X_i, \theta)$, $\quad \ell(\theta) = \sum \log f(X_i, \theta)$

Set $\ell'(\theta) = 0$, verify $\ell''(\hat{\theta}) < 0$.

Dist	MLE
Ber(θ), Poi(θ)	$\hat{\theta} = \bar{X}$
$N(\mu, \sigma^2)$	$\hat{\mu} = \bar{X}$, $\hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$ **(BIASED)**
Exp(θ) rate	$\hat{\theta} = 1/\bar{X}$
Unif$[0, \theta]$	$\hat{\theta} = X_{(n)} = \max(X_i)$ (by inspection)
Unif$[\theta, \theta{+}1]$	any $\theta \in [X_{(n)} - 1,\ X_{(1)}]$ (NOT unique)

Properties:

- **Invariance**: $g(\hat{\theta})$ is MLE of $g(\theta)$ for any function g
- **Consistent**: $\hat{\theta}_n \xrightarrow{P} \theta$ (under regularity conditions)
- **Caution**: MLE may NOT exist, NOT be unique, NOT be unbiased

When calculus fails: Unif$[0, \theta]$: $L(\theta) = \theta^{-n}\, I(\theta \geq X_{(n)})$ — decreasing in θ, so max at $\hat{\theta} = X_{(n)}$.

MSE & Comparing Estimators

$$\text{MSE}(T_n) = \text{Var}(T_n) + [\text{bias}(T_n)]^2$$

For unbiased estimator: MSE = Var.

Estimator	Unbiased?	MVUE?	Notes
$\bar{X}$ for μ	Yes (always)	Yes (Normal)	Var $= \sigma^2/n$
$\frac{1}{n}\sum(X_i - \bar{X})^2$	No (for σ^2)	No	$E = \frac{n-1}{n}\sigma^2$
$s^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$	Yes (σ^2)	Yes (Normal)	Bessel's
$2\bar{X}$ for Unif$[0, \theta]$	Yes	No	MSE $= \theta^2/(3n)$
$X_{(n)}$ for Unif$[0, \theta]$	**No**	No	MSE $\ll 2\bar{X}$
$(1{+}1/n)X_{(n)}$	Yes	Yes (MVUE)	MSE $= \frac{\theta^2}{n(n+2)}$

Biased can beat unbiased (bias-variance tradeoff).

Unbiased estimators don't always exist: $\theta/(1-\theta)$ for Ber(θ) has none.

No universally best estimator in general.

Sufficient Statistics

T is sufficient for $\theta \;\Leftrightarrow\; f(\mathbf{x}, \theta) = u(\mathbf{x}) \cdot v(r(\mathbf{x}), \theta)$ **(Factorization Criterion)**

- $u(\mathbf{x})$: depends on data ONLY (not θ)
- $v(r(\mathbf{x}), \theta)$: θ enters ONLY through $r(\mathbf{x})$

Distribution	Sufficient Statistic
Poi(θ), Ber(θ), Exp(θ)	$\sum X_i$
$N(\mu, \sigma^2)$ both unknown	$(\sum X_i,\ \sum X_i^2)$
$N(\mu, \sigma^2$ known)	$\sum X_i$
Unif$[0, \theta]$	$X_{(n)} = \max(X_i)$
Gam(α known, β)	$\sum X_i$
Gam(α, β known)	$\prod X_i$

Rao-Blackwell Theorem

Given ANY estimator $\delta(X)$ and sufficient statistic T:

$$\delta_0(T) = E[\delta(X) \mid T]$$

Then: MSE$(\delta_0) \leq$ MSE(δ) for all θ (never worse, often strictly better).

$\Rightarrow$ Always condition on sufficient statistics to improve estimators.

Bayesian Inference

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}: \quad \xi(\theta|x) \propto f(x|\theta) \cdot \xi(\theta)$$

Bayes est (squared error) $= E[\theta|X]$ (posterior mean).

Bayes est (absolute error) = posterior median.

Conjugate Priors

Likelihood	Conjugate Prior	Posterior
Ber(θ)	Beta(α, β)	Beta($\alpha{+}\sum x_i,\ \beta{+}n{-}\sum x_i$)
Poi(λ)	Gam(α, β)	Gam($\alpha{+}\sum x_i,\ \beta{+}n$)
$N(\mu, \sigma^2$ known)	$N(\mu_0, v_0^2)$	$N(\mu_n, v_n^2)$
Exp(θ)	Gam(α, β)	Gam($\alpha{+}n,\ \beta{+}\sum x_i$)

Normal–Normal Conjugate (σ^2 known)

Prior: $\mu \sim N(\mu_0, v_0^2)$. Data: $X_1, \ldots, X_n$ iid $N(\mu, \sigma^2)$.

Posterior: $\mu|X \sim N(\mu_n, v_n^2)$ where:

$$\mu_n = \frac{\mu_0/v_0^2 + n\bar{X}/\sigma^2}{1/v_0^2 + n/\sigma^2}, \qquad v_n^2 = \frac{1}{1/v_0^2 + n/\sigma^2}$$

- Posterior mean = weighted average of prior mean and $\bar{X}$, weighted by **precisions** (1/variance)
- As $n \to \infty$: posterior concentrates around $\bar{X}$ (data overwhelms prior)

- As $v_0^2 \to \infty$ (vague prior): posterior $\approx N(\bar{X}, \sigma^2/n)$

Sampling Distributions

Chi-Squared

$Z \sim N(0,1) \;\Rightarrow\; Z^2 \sim \chi_1^2$.
$Z_1, \ldots, Z_d$ iid $N(0,1) \;\Rightarrow\; \sum Z_i^2 \sim \chi_d^2$.
$\chi_d^2 = \text{Gam}(d/2,\ 1/2)$. $\quad E(\chi_d^2) = d, \quad \text{Var}(\chi_d^2) = 2d$.
Reproductive: $\sum \chi_{m_i}^2 \sim \chi_{\sum m_i}^2$ (independent).

Normal Sampling — CRITICAL RESULTS

$X_1, \ldots, X_n$ iid $N(\mu, \sigma^2)$:

1. $\bar{X} \sim N(\mu,\ \sigma^2/n)$
2. $s^2 \sim \dfrac{\sigma^2}{n-1}\chi_{n-1}^2$, equivalently $\dfrac{S^2}{\sigma^2} = \dfrac{\sum(X_i - \bar{X})^2}{\sigma^2} \sim \chi_{n-1}^2$
3. $\bar{X}$ and s^2 are **INDEPENDENT** (Proposition 6.3)

Independence of $\bar{X}$ and s^2 **characterizes** normality.

Theorem 6.4 (statement only): Z_i iid $N(0,1)$, A orthogonal $\Rightarrow$ $V = AZ$ has V_i iid $N(0,1)$ and $\sum V_i^2 = \sum Z_i^2$.

t-Distribution

$Z \sim N(0,1)$, $V \sim \chi_n^2$, $Z \perp V$:

$$T = \frac{Z}{\sqrt{V/n}} \sim t_n$$

Properties:

- Symmetric about 0 (T and $-T$ have same distribution)
- Heavier tails than $N(0,1)$ for finite n
- As $n \to \infty$: $t_n \to N(0,1)$ (by WLLN: $V/n \xrightarrow{P} 1$)

Theorem 6.5: X_i iid $N(\mu, \sigma^2)$:

$$\frac{\sqrt{n}(\bar{X} - \mu)}{s} \sim t_{n-1}$$

Confidence Intervals

(A, B) is level $1-\alpha$ CI if $P_\theta(A \le g(\theta) \le B) \ge 1-\alpha$ for all θ.

Method of Pivots

1. Construct $\Psi(\text{data}, \theta)$ with known distribution FREE of θ
2. Find quantiles: $P(q_{\beta_1} \le \Psi \le q_{1-\beta_2}) = 1-\alpha$ where $\beta_1 + \beta_2 = \alpha$
3. Invert the inequalities to isolate $g(\theta)$

Equal-tailed ($\beta_1 = \beta_2 = \alpha/2$) gives shortest CI for symmetric distributions.

Normal Mean CIs

Case	Pivot	Dist	CI
σ^2 known	$\frac{\sqrt{n}(\bar{X}-\mu)}{\sigma}$	$N(0,1)$	$\bar{X} \pm \frac{\sigma}{\sqrt{n}} z_{\alpha/2}$
σ^2 unknown	$\frac{\sqrt{n}(\bar{X}-\mu)}{s}$	t_{n-1}	$\bar{X} \pm \frac{s}{\sqrt{n}} t_{n-1,\alpha/2}$
Large n, any dist	$\frac{\sqrt{n}(\bar{X}-\mu)}{s}$	$\approx N(0,1)$	$\bar{X} \pm \frac{s}{\sqrt{n}} z_{\alpha/2}$

Other CIs

Bernoulli (large n): $\hat{\theta} \pm \sqrt{\dfrac{\hat{\theta}(1-\hat{\theta})}{n-1}}\, z_{\alpha/2}$

CI for σ^2 (normal data): Pivot $(n-1)s^2/\sigma^2 \sim \chi_{n-1}^2$.

$$\text{CI}: \quad \left(\frac{(n-1)s^2}{\chi_{n-1,\alpha/2}^2}, \quad \frac{(n-1)s^2}{\chi_{n-1,1-\alpha/2}^2} \right)$$

CI Properties

- Width (normal, σ known): $2\sigma\, z_{\alpha/2}/\sqrt{n}$
- Higher confidence (smaller α) $\to$ WIDER CI
- Larger $n \to$ NARROWER CI (width $\propto 1/\sqrt{n}$)
- $4\times$ sample size $\Rightarrow$ half the CI width
- $\alpha = 0 \Rightarrow z_{\alpha/2} = \infty \Rightarrow$ infinitely wide

CI Interpretation (Classic T/F Trap!)

CORRECT: "We have *confidence* γ that θ lies in (a, b)."

WRONG: "θ lies in (a, b) with *probability* γ." — θ is FIXED (not random), the interval is random. After observing data, it either contains θ or it doesn't.

Cramér-Rao Inequality

Score: $\dot{\ell}(x, \theta) = \frac{\partial}{\partial\theta} \log f(x, \theta)$. $\quad E[\dot{\ell}] = 0$. $\quad$ Fisher info: $I(\theta) = E[\dot{\ell}^2]$.
For n iid observations: total info $= n\, I(\theta)$.

Regularity Conditions (KNOW THESE)

(A.1) Support $\{x : f(x, \theta) > 0\}$ does NOT depend on θ.
Excludes: Unif$[0, \theta]$, Unif$[\theta, \theta+1]$ — support changes with θ.

(A.2) Can interchange $\frac{\partial}{\partial\theta}$ and $\int$:

$$\frac{\partial}{\partial\theta} \int W(\mathbf{x})\, f(\mathbf{x}, \theta)\, d\mathbf{x} = \int W(\mathbf{x})\, \frac{\partial f}{\partial\theta}(\mathbf{x}, \theta)\, d\mathbf{x}$$

(A.3) $\frac{\partial}{\partial\theta} \log f(x, \theta)$ exists for all $x \in A$ and all $\theta \in \Omega$, as a finite quantity.

The CRLB

If $T(\mathbf{X}_n)$ is UNBIASED for $g(\theta)$, and A.1–A.3 hold, and $I(\theta) < \infty$:

$$\boxed{\text{Var}_\theta(T) \;\ge\; \frac{[g'(\theta)]^2}{n\, I(\theta)}}$$

Special case $g(\theta) = \theta$: $\quad \text{Var}(T) \ge 1/(n\, I(\theta))$.
Achieves the bound $\Rightarrow$ **MVUE** (efficient estimator).

Common Fisher Information

Dist	$f(x, \theta)$	$\dot{\ell}(x, \theta)$	$I(\theta)$	CRLB
Ber(θ)	$\theta^x(1-\theta)^{1-x}$	$\frac{x-\theta}{\theta(1-\theta)}$	$\frac{1}{\theta(1-\theta)}$	$\frac{\theta(1-\theta)}{n}$
Poi(θ)	$e^{-\theta}\theta^x/x!$	$x/\theta - 1$	$1/\theta$	θ/n
$N(\mu, \sigma^2)$	normal pdf	$(x-\mu)/\sigma^2$	$1/\sigma^2$	σ^2/n
Exp(θ)	$\theta e^{-\theta x}$	$1/\theta - x$	$1/\theta^2$	θ^2/n

$\bar{X}$ is MVUE for: μ in Normal, θ in Bernoulli, θ in Poisson — all achieve CRLB.

True/False Traps (by Topic)

Convergence: **F**: conv in dist $\Rightarrow$ conv in prob (only if limit is constant). **F**: CLT says $\bar{X}$ is exactly normal (approx only). **T**: conv in prob $\Rightarrow$ conv in dist.

MLE: **F**: MLE always unbiased ($\hat{\sigma}^2$, $X_{(n)}$ biased). **F**: MLE always exists/unique. **T**: MLE invariant: $g(\hat{\theta})$ is MLE of $g(\theta)$. **T**: MLE consistent (reg. conds). **T**: MLE asymp. normal: $\hat{\theta}_n \approx N(\theta, 1/(nI(\theta)))$.

Estimators: **F**: unbiased always better (biased $X_{(n)} \ll 2\bar{X}$ for Unif). **F**: consistent $\Rightarrow$ unbiased. **F**: unbiased est always exists ($\theta/(1-\theta)$ for Ber: none). **T**: biased can have lower MSE than any unbiased.

Sufficiency & Info: **T**: 1-to-1 function of suff. stat is sufficient. **T**: suff stat can be vector-valued. **T**: Rao-Blackwell improves via sufficiency. **F**: CRLB applies to all dists (fails for Uniform: A.1).

CI: **F**: "$\theta \in$ CI with prob $1-\alpha$" (θ fixed; say confidence).

Normal sampling: **T**: $\bar{X} \perp s^2$ (Prop 6.3). **T**: $t_n \to N(0,1)$.

Bayesian: **T**: posterior $\to$ true θ as $n \to \infty$ regardless of prior.

Fill-in-the-Blank Quick Lookup

$Z_n \xrightarrow{P} c$, g cts $\Rightarrow g(Z_n) \xrightarrow{P} g(c)$	Continuous Mapping Thm
$X_n \xrightarrow{d} X$, $Y_n \xrightarrow{P} a \Rightarrow X_n Y_n \xrightarrow{d} aX$	Slutsky's Theorem
Minimizes posterior expected loss	Bayes estimator
Sq. error $\to$ post. mean; abs. error $\to$ post. median	
$E[\dot{\ell}(X,\theta)]$	$= 0$ (score has mean 0)
$I(\theta) = E[\dot{\ell}^2] = \mathrm{Var}[\dot{\ell}]$	Fisher information

Computation Recipes

1. **Find MLE**: Write $L(\theta)$, take log, $\ell'(\theta) = 0$, check $\ell'' < 0$. Uniform: by inspection.
2. **Find MOM**: Set $E(X^j) = \frac{1}{n}\sum X_i^j$, solve for parameters.
3. **Unbiased?**: Compute $E(\text{est})$, check if $= g(\theta)$ for ALL θ.
4. **Sufficient stat**: Write joint pdf, factor as $u(\mathbf{x}) \cdot v(r(\mathbf{x}), \theta)$.
5. **Compare estimators**: MSE = Var + Bias^2 for each, compare across all θ.
6. **Show consistency**: Show $\hat{\theta}_n \xrightarrow{P} \theta$ via LLN, Chebyshev, or CMT.
7. **Asymptotic dist**: CLT for $\bar{X}$, then delta method if transformed.
8. **Construct CI**: Pick pivot (Z, t, or χ^2), find quantiles, invert.
9. **Fisher information**: $\dot{\ell} = \partial/\partial\theta \log f$, then $I(\theta) = E[\dot{\ell}^2]$.
10. **Is it MVUE?**: Check unbiased + Var = CRLB, or use Rao-Blackwell.
11. **Find posterior**: Write $\propto$ likelihood $\times$ prior, identify the distribution kernel.
12. **Bayes estimator**: Sq. error = posterior mean; abs. error = posterior median.
13. **Interpret CI**: Say "confidence," NOT "probability." θ is fixed.
14. **Rao-Blackwell**: Find sufficient T, compute $E[\delta(X)|T]$.
15. **Sample size**: Use Chebyshev or CI width formula. $n \geq \sigma^2/(\varepsilon^2\delta)$.

Workflow Guides (Plug-and-Play)

Quick Classifier

"Find $\mathrm{Var}(aX + bY + \cdots)$"	$\to$ **W1**
"Bound on $P(\ldots)$" or "How large n?"	$\to$ **W2**
"Asymptotic dist of $g(\bar{X})$"	$\to$ **W3**
"CDF/dist of $X_{(n)} = \max(X_i)$"	$\to$ **W4**
"Find the MLE of θ"	$\to$ **W5**
"Find MLE of $g(\theta)$"	$\to$ **W6**
"Find MOM" or "Show MOM = MLE"	$\to$ **W7**
"Show $\hat{\theta}$ is consistent"	$\to$ **W8**
"Evaluate $\int x^a e^{-bx} dx$" or "$E[X^k]$"	$\to$ **W9**
"Dist of $\sum X_i$" or "Find MGF"	$\to$ **W10**
Fisher info $\to$ efficiency $\to$ asymp MLE $\to$ CI	$\to$ **W11**
"Margin of error $1/k$ as large"	$\to$ **W12**
"Show T sufficient via conditional dist"	$\to$ **W13**

W1: Variance of Linear Combinations

Trigger: Find $\mathrm{Var}(a_1X_1 + a_2X_2 + \cdots + c)$

Formula:

$$\mathrm{Var}\Big(\sum a_i X_i + c\Big) = \sum_i a_i^2 \mathrm{Var}(X_i) + 2\sum_{i<j} a_i a_j \mathrm{Cov}(X_i, X_j)$$

Steps: (1) Drop constants. (2) List coefficients a_i *with signs*. (3) Square each $\times$ Var. (4) Cross terms: $2a_ia_j$Cov. (5) Sum.

Check: Var ≥ 0. Negative $\Rightarrow$ sign error in cross terms.

W2: Probability Bounds & Sample Size

Markov ($W \geq 0$): $P(W \geq t) \leq E(W)/t$. Use when only mean known.

Chebyshev: $P(|Z - EZ| \geq \tau) \leq \mathrm{Var}(Z)/\tau^2$. Use when variance known.

Higher moment: $P(|X-\mu| \geq t) \leq E|X-\mu|^k/t^k$. Use when β_k given.

Sample size via Chebyshev: Want $P(|\bar{X} - \mu| \geq \varepsilon) \leq \delta$.

$$\frac{\sigma^2}{n\varepsilon^2} \leq \delta \implies \boxed{n \geq \frac{\sigma^2}{\varepsilon^2\delta}}$$

Conv in prob via Markov: Show $E(X_n) \to 0 \Rightarrow P(X_n \geq \varepsilon) \leq E(X_n)/\varepsilon \to 0$.

W3: Delta Method

Trigger: Asymptotic distribution of $g(Z_n)$ where $\sqrt{n}(Z_n - \theta)/\sigma \xrightarrow{d} N(0,1)$.

Standard ($g'(\theta) \neq 0$):

1. Identify Z_n, its asymptotic mean θ, and asymptotic variance σ^2/n
2. Compute $g'(\theta)$; verify $g'(\theta) \neq 0$
3. Answer: $g(Z_n) \sim N\big(g(\theta),\ (\sigma^2/n)[g'(\theta)]^2\big)$

Second-order ($g'(\theta) = 0$): Standard gives degenerate limit.

If $\sqrt{n}\, Z_n \xrightarrow{d} N(0, \sigma^2)$, then $n\, Z_n^2 = (\sqrt{n}\, Z_n)^2 \xrightarrow{d} \sigma^2\chi_1^2$ by CMT.

More generally: $n(g(Z_n) - g(\theta)) \xrightarrow{d} \frac{\sigma^2 g''(\theta)}{2}\chi_1^2$.

W4: Order Statistics

CDF of max: $P(X_{(n)} \leq y) = [F_X(y)]^n$

Convergence: Let $Z_n = n(X_{(n)} - \theta)$.

$$P(Z_n \leq z) = \big[F_X\big(\theta + \tfrac{z}{n}\big)\big]^n \xrightarrow{n\to\infty} e^{z\, f(\theta)}$$

(using $(1 + a/n)^n \to e^a$). Rate is n, not $\sqrt{n}$.

W5: MLE Computation

Step 0: Is support θ-dependent? Yes $\to$ inspection. No $\to$ calculus.

Calculus path — step by step:

1. Likelihood: $L(\theta) = \prod f(x_i|\theta)$. Key: $\prod \theta = \theta^n$, $\prod e^{-\theta x_i} = e^{-\theta \sum x_i}$.

2. Log-likelihood: $\ell(\theta) = \sum \log f(x_i|\theta)$. Drop terms not involving θ.

Key: $\log(\theta^n) = n\log\theta$, $\log(e^{-n\theta}) = -n\theta$.

3. Differentiate: $\frac{d}{d\theta}[n\log\theta] = n/\theta$, $\frac{d}{d\theta}[-n\theta] = -n$, $\frac{d}{d\theta}[(\sum x_i)\log\theta] = \sum x_i/\theta$.

4. Set $\ell'(\theta) = 0$, solve: Pattern is $n/\theta - (\text{sum}) = 0$.

5. Verify: $\ell''(\hat{\theta}) < 0$ (common: $-n/\theta^2 < 0$ ✓).

Exp(β): $\ell = n\log\beta - \beta\sum x_i$. $\ell' = n/\beta - \sum x_i = 0$. $\hat{\beta} = 1/\bar{X}$.

Ber(θ): $\ell = (\sum x_i)\log\theta + (n - \sum x_i)\log(1-\theta)$. $\ell' = \sum x_i/\theta - (n - \sum x_i)/(1-\theta) = 0$. $\hat{\theta} = \bar{X}$.

Power $f = \theta x^{\theta-1}$: $\ell = n\log\theta + (\theta-1)\sum\log x_i$. $\ell' = n/\theta + \sum\log x_i = 0$. $\hat{\theta} = -n/\sum\log x_i$.

Inspection (Uniform): $L(\theta) = \theta^{-n} I(\theta \geq X_{(n)})$, decreasing $\Rightarrow \hat{\theta} = X_{(n)}$.

Restricted domain: Find unrestricted MLE. If outside domain, pick closest boundary.

W6: MLE Invariance

Rule: $\hat{\theta}$ is MLE of $\theta \Rightarrow g(\hat{\theta})$ is MLE of $g(\theta)$.

Just plug $\hat{\theta}$ into g.

Examples: MLE of median $= (\log 2)/\hat{\beta}$. MLE of $\theta^2 = \hat{\theta}^2$. MLE of $P(X > c) = e^{-c\hat{\beta}}$.

W7: MOM (& MOM = MLE)

1. Population moments: $\mu_1 = E(X)$, $\mu_2 = E(X^2)$
2. Sample moments: $\hat{\mu}_1 = \bar{X}$, $\hat{\mu}_2 = \frac{1}{n}\sum X_i^2$
3. Set $\mu_k(\theta) = \hat{\mu}_k$, solve

MOM = MLE for: Ber, Poi, Exp, Normal. NOT for Gamma (MLE needs numerics).

W8: Consistency Proofs

Given: estimator $\hat{\theta}_n$ and distribution (so you know $E(X)$). **Write 3 lines**:

LLN + CMT template (use this 90% of the time):

Line 1: "By WLLN, $\bar{X}_n \xrightarrow{P} E(X) = [\text{mean}]$."

Line 2: "Let $g(x) = [\text{function s.t. } g(\bar{X}) = \hat{\theta}]$. g is continuous at $x = [\text{mean}]$."

Line 3: "By CMT, $g(\bar{X}_n) \xrightarrow{P} g(\text{mean}) = \theta$. $\square$"

How to find g: $\hat{\theta} = 1/\bar{X} \Rightarrow g(x) = 1/x$. $\hat{\theta} = \bar{X}^2 \Rightarrow g(x) = x^2$.

Exp(β), $\hat{\beta} = 1/\bar{X}$: WLLN $\bar{X} \xrightarrow{P} 1/\beta$; $g(x) = 1/x$ cts; CMT $\Rightarrow 1/\bar{X} \xrightarrow{P} \beta$.

Any dist, $\hat{\theta} = \bar{X}^2$: WLLN $\bar{X} \xrightarrow{P} \mu$; $g(x) = x^2$ cts; CMT $\Rightarrow \bar{X}^2 \xrightarrow{P} \mu^2$.

MSE method (when they give you $\text{Var}(\hat{\theta})$): show bias $\to 0$ and Var $\to 0$, then MSE $\to 0 \Rightarrow \hat{\theta}_n \xrightarrow{P} \theta$.

W9: Gamma Integrals & $E[X^k]$

Master formula: $\displaystyle\int_0^\infty x^{\alpha-1}e^{-\beta x}\,dx = \frac{\Gamma(\alpha)}{\beta^\alpha}$

Match your integral to this form. Read off α = power + 1, β = coeff in exponent.

$E[X^k]$ **for** $X \sim$ **Gam**(α, β):

$$E[X^k] = \frac{\Gamma(\alpha+k)}{\beta^k\,\Gamma(\alpha)} = \frac{\alpha(\alpha+1)\cdots(\alpha+k-1)}{\beta^k}$$

Key: $\Gamma(n) = (n-1)!$, $\Gamma(1/2) = \sqrt{\pi}$, $\Gamma(\alpha+1) = \alpha\,\Gamma(\alpha)$.

W10: MGF Products & Sampling Dists

Sum of independents: $M_{\sum X_i}(t) = \prod M_{X_i}(t)$. For iid: $[M_X(t)]^n$.

Recognize MGF: $(1-p+pe^t)^n \to$ Bin, $e^{\lambda(e^t-1)} \to$ Poi, $(\beta/(\beta-t))^\alpha \to$ Gam

Building a t**-statistic**:

1. Find $Z \sim N(0,1)$ (normalized sum)
2. Find independent $V \sim \chi^2_m$ (sum of squared normals)
3. $T = Z/\sqrt{V/m} \sim t_m$

Key facts: $Z^2 \sim \chi^2_1$. $(n-1)s^2/\sigma^2 \sim \chi^2_{n-1}$. $\sqrt{n}(\bar{X}-\mu)/s \sim t_{n-1}$.

W11: Full Estimation Pipeline

Given: distribution $f(x|\theta)$, sample size n. θ **and** n **stay variables.**

(a) Fisher info — compute score, square it, take expectation:

$\log f \to \dot{\ell} = \partial/\partial\theta \log f \to I_1(\theta) = E[\dot{\ell}^2] = \text{Var}(\dot{\ell})$

$I_n(\theta) = n\,I_1(\theta)$

Dist	$\dot{\ell}$	$I_1(\theta)$	$I_n(\theta)$
Poi(θ)	$(x-\theta)/\theta$	$1/\theta$	n/θ
Exp(θ)	$1/\theta - x$	$1/\theta^2$	n/θ^2
Ber(θ)	$(x-\theta)/[\theta(1-\theta)]$	$1/[\theta(1-\theta)]$	$n/[\theta(1-\theta)]$
$N(\mu,\sigma^2)$	$(x-\mu)/\sigma^2$	$1/\sigma^2$	n/σ^2

(b) Efficiency: Exam gives T and $\text{Var}(T)$. Compare to CRLB $= 1/(nI_1)$.

$\text{Var}(T) = \text{CRLB} \Rightarrow$ efficient. $\text{Var}(T) > \text{CRLB} \Rightarrow$ not efficient.

(c) Asymp MLE: $\hat{\theta}_n \sim_{\text{approx}} N(\theta,\ 1/(nI_1(\theta)))$. Just plug in I_1.

Poi: $N(\theta, \theta/n)$. Exp: $N(\theta, \theta^2/n)$. Ber: $N(\theta, \theta(1-\theta)/n)$.

(d) CI: Plug $\hat{\theta}$ for θ: $\hat{\theta} \pm z_{\alpha/2}\sqrt{1/(n\,I_1(\hat{\theta}))}$.

Exp: $1/\bar{X} \pm z_{\alpha/2} \cdot (1/\bar{X})/\sqrt{n}$. Poi: $\bar{X} \pm z_{\alpha/2}\sqrt{\bar{X}/n}$.

W12: CI Width Scaling

CI width $\propto 1/\sqrt{n}$. To cut margin of error by factor k: $\boxed{n_{\text{new}} = k^2 \cdot n_{\text{old}}}$

Ex: margin 1/3 as large $\Rightarrow n_{\text{new}} = 9 \cdot n_{\text{old}}$.

Sample size for target margin E (proportion): $n \geq z^2_{\alpha/2}\,\hat{p}(1-\hat{p})/E^2$.

Conservative ($\hat{p} = 0.5$): $n \geq z^2_{\alpha/2}/(4E^2)$.

Via Chebyshev: $P(|\bar{X} - \mu| \leq k\sigma) \geq 1 - \delta \Rightarrow n \geq 1/(k^2\delta)$.

W13: Sufficiency (Two Methods)

Given: distribution $f(x|\theta)$ and statistic T. **You need**: distribution of T for Method B.

Method A: Factorization (default, faster):

Write joint $f(\mathbf{x};\theta) = \underbrace{[\text{no } \theta]}_{u(\mathbf{x})} \cdot \underbrace{[\theta \text{ only thru } T]}_{v(T,\theta)}$. Done.

Poi: $\frac{e^{-n\theta}\theta^{\sum x_i}}{\prod x_i!} = \frac{1}{\prod x_i!} \cdot e^{-n\theta}\theta^{\sum x_i}$. $T = \sum X_i$. ✓

Exp: $\theta^n e^{-\theta\sum x_i} = 1 \cdot \theta^n e^{-\theta\sum x_i}$. $T = \sum X_i$. ✓

$N(\mu, \sigma^2$ known$)$: $\propto \exp\left(\frac{\mu\sum x_i}{\sigma^2} - \frac{n\mu^2}{2\sigma^2}\right) \cdot \exp\left(\frac{-\sum x_i^2}{2\sigma^2}\right)$. $T = \sum X_i$. ✓

Method B: Conditional (when problem says "using the definition"):

Compute $f(\mathbf{x}|T{=}t) = f(\mathbf{x};\theta)\,/\,P(T{=}t;\theta)$. All θ must cancel.

Need to know $P(T{=}t)$: Ber $\Rightarrow T \sim \text{Bin}(n,\theta)$; Poi $\Rightarrow T \sim \text{Poi}(n\theta)$.

Ber: $\frac{\theta^t(1-\theta)^{n-t}}{\binom{n}{t}\theta^t(1-\theta)^{n-t}} = \frac{1}{\binom{n}{t}}$. No θ. ✓

Poi: $\frac{e^{-n\theta}\theta^t/\prod x_i!}{e^{-n\theta}(n\theta)^t/t!} = \frac{t!}{n^t\prod x_i!}$. No θ. ✓